

Moosa Abbasi

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Education

University of South Florida, Honors College

Bachelor of Science in Computer Science

GPA: 3.90/4.00 | Green and Gold Presidential Scholar, Dean's List

Tampa, FL

Aug. 2023 – Aug. 2027

Relevant Coursework: Data Structures, Discrete Structures, Analysis of Algorithms, Computer Architecture, Software Engineering

Technical Skills

Languages & Databases: Python, C, C++, JavaScript, SQL, DynamoDB, MySQL

Frameworks & Libraries: Streamlit, React.js, OpenCV, Pandas, NumPy, Plotly, Flask

Cloud & DevOps: AWS (Lambda, API Gateway, S3, CloudWatch, IAM), Docker, Git, GitHub Actions

Security & Testing: Burp Suite, OWASP ZAP, OWASP WSTG v4.2, HTTP Traffic Analysis, XSS/SQLi, Prompt Injection

Certifications: Machine Learning Specialization (Supervised ML, Advanced Learning Algorithms, Unsupervised Learning)—Andrew Ng, DeepLearning.AI & Stanford University, Coursera · 2026

Experience

Undergraduate Research Assistant — AI Course Companion (AICC)

May 2026 – Present

University of South Florida | CIS 4915 — Dr. Oguzhan Topsakal

- Identified gap in **OWASP WSTG v4.2** coverage for **LLM-specific** attacks; designed a **27-case black-box** security plan for **AICC** across **5 phases**, independently adding **prompt injection** as a supplementary attack category not yet covered in the standard.
- Executing the plan using **Burp Suite** and **OWASP ZAP**, conducting **HTTP traffic interception**, **session token analysis**, **XSS/SQLi** payload testing, **CSRF** evaluation, and **prompt injection** attacks with severity-rated findings documented for remediation.

Project Management Intern

May 2024 – Aug 2024

Huawei Technologies

- Resolved visibility gaps across **5+ concurrent DMD projects** by building structured **tracking dashboards** that standardized reporting cycles and cross-team workflows, improving on-time delivery by **20%**.
- Eliminated siloed communication by introducing a **data-driven reporting** system and standardized **project documentation** workflow, reducing delays by **25%** and boosting team efficiency by **15%**.

Projects

Smart Job Tracker

May 2025 – Present

AWS Lambda, DynamoDB, API Gateway, Cognito, SES, Chrome Extension (MV3), Streamlit, Python

Tampa, FL

- Eliminated manual logging by building a **Chrome Extension (MV3)** with a **3-layer detection system** using **MAIN world XHR monkey-patching** and **MutationObserver**; implemented a **chrome.alarms** job queue guaranteeing **zero application loss** during service worker suspension.
- Engineered a **5-Lambda serverless backend** with **DynamoDB user-scoped isolation**, **API Gateway REST API** protected by a **Cognito JWT** authorizer, **preemptive token refresh** with race-condition locking, and automatic **401 retry** across all 3 clients.
- Deployed a live **Streamlit** dashboard with **EventBridge + SES** daily email reminders for application follow-up tracking; reduced total operating cost to **\$0/month** using the **AWS Free Tier**.

USF Emissions Analysis Dashboard

Jan 2026 – Present

Python, Streamlit, Plotly, Pandas, GHG Protocol

Tampa, FL

- Quantified **~83.8M kg CO₂** across **5 transportation categories** (Scope 1 and Scope 3 Cat 4/6/7/9) by building a full-stack **GHG emissions dashboard** in **Python/Streamlit** with **GHG Protocol**-aligned emission factor calculations across **~10,000 synthetic records**.
- Eliminated reliance on third-party tools by implementing **EPA SmartWay** and **GHG Protocol Corporate Standard** emission factor formulas from scratch for **5 datasets**, fleet, logistics, business travel, commuting, and downstream transport.
- Delivered findings to an industry airport client by deploying on **Streamlit Cloud** via **GitHub** with automated redeployment; collaborated with **Patel's College of Global Sustainability** for Scope 1 and Scope 3 strategy recommendations.

Digital Image Processing (DIP) System

Jan 2025 – Apr 2025

Python, OpenCV, NumPy, Matplotlib, Image Segmentation

Tampa, FL

- Improved edge detection accuracy by **35%** and visual precision by **40%** across **500+ datasets** by engineering a preprocessing pipeline using **Python**, **OpenCV**, **Gaussian/Sobel** filtering, and adaptive **Otsu thresholding** algorithms.
- Reduced processing latency by **30%** by building a modular real-time analysis pipeline with **Matplotlib** visualization and automated benchmarking, enabling efficient model evaluation and performance debugging.
- Evaluated **5+ segmentation techniques**, including **K-means clustering**, **Watershed**, and **morphological operations** against ground truth masks, quantifying boundary detection quality improvements across varied noise and lighting conditions.